

Exposé VII, continuation: compatibilities and trivializations

3.7. Various compatibilities

The preceding discussion first gives, for variable commutative Groups, an equivalence between central extensions of a commutative Group P by variable commutative Groups G , and the cofibred category $\Psi_{K(P)}$, where $K(P)$ denotes the complex attached to P . Similarly, for two Groups P, Q , the cofibred category of biextensions of (P, Q) is equivalent to the cofibred category

$$\Psi_{L(P,Q)}, \quad L(P, Q) = K(P) \otimes K(Q).$$

These remarks suggest that, for P commutative, one may perhaps find a canonical resolution of P of the kind desired in 3.5.4(a), by a judicious modification of the chain complex $C(P, \mathbb{Z})$.

In the present section we examine the behaviour of the equivalence of cofibred categories 3.6.4 when P and Q vary. The variance of the cofibred category of biextensions of (P, Q) by variable G , with respect to the variables P, Q , was defined in principle in 2.4, and may also be formulated by saying that one has a cofibred functor (2.6.1). Notice that the variance in P, Q is contravariant. The variance of the cofibred category $\Psi_{L(P,Q)}$, with respect to P, Q , is equally clear, because $L(P, Q) = L(P) \otimes L(Q)$ depends functorially on the pair (P, Q) , whereas Ψ_L depends contravariantly on L (3.4). Thus a first compatibility says that, for every triple of morphisms

$$P' \longrightarrow P, \quad Q' \longrightarrow Q, \quad G \longrightarrow G', \quad (3.7.1)$$

one has a commutative square of functors, up to the canonical isomorphism,

$$\begin{array}{ccc} \Psi_{L(P,Q)}(G) & \longrightarrow & \Psi_{L(P',Q')}(G') \\ \downarrow & & \downarrow \\ \text{Biext}(P, Q; G) & \longrightarrow & \text{Biext}(P', Q'; G'). \end{array} \quad (3.7.2)$$

We shall admit this compatibility, since writing down the commutativity isomorphism is no less tedious than evident; it has the same character as the complete explicit construction involved in the equivalence of cofibred categories 3.6.4.

It follows in particular that the isomorphism (3.6.5.1) is functorial not only in G , but also in P and Q . Similarly, and more trivially, the isomorphism defined through (3.1.8), as (3.6.5.1) was defined through (3.2.4), gives

$$\text{Biext}^0(P, Q; G) \xrightarrow{\sim} \text{Ext}^0(P \otimes^L Q, G) \simeq \text{Hom}(P \otimes Q, G). \quad (3.7.4)$$

This is the same isomorphism already obtained in (2.5.4.1), where it was noted in (2.6) that it is functorial in the three arguments.

The interest of (3.6.5) is that it expresses the geometrically defined group $\text{Biext}^1(P, Q; G)$ by the simple homological invariant $\text{Ext}^1(P \otimes^L Q, G)$, which belongs to the usual sequence of invariants $\text{Ext}^i(P \otimes^L Q, G)$. The latter sequence is a cohomological functor in each of the three arguments. Thus a short exact sequence

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \quad (3.7.5.1)$$

or, respectively,

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0, \quad \text{or} \quad 0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0, \quad (3.7.5.2)$$

gives rise to an infinite exact sequence in the Ext^i 's. Its first six terms, involving only Ext^0 's and Ext^1 's, may be interpreted in terms of biextensions by 3.6.5 and 3.7.4. We have already indicated the geometric interpretation of the exact sequence deduced from (3.7.5.1), and in particular of the boundary operator

$$\partial : \text{Biext}^0(P, Q; G'') \longrightarrow \text{Biext}^1(P, Q; G'), \quad (3.7.5.3)$$

in the framework of left exact cofibred categories (3.3.6). One may likewise ask for a geometric interpretation of the six-term exact sequence attached to either of the short exact sequences (3.7.5.2), say the first; equivalently, by 3.7.3, one seeks an interpretation of the boundary homomorphism

$$\partial : \text{Biext}^0(P', Q; G) \longrightarrow \text{Biext}^1(P'', Q; G). \quad (3.7.5.4)$$

Proposition 3.7.6. The cofibred category, over $\text{GRAB}(\mathcal{T})^\circ \times \text{GRAB}(\mathcal{T})^\circ \times \text{GRAB}(\mathcal{T})$, of biextensions of (P, Q) by G , with P, Q, G variable, is left exact not only in the argument G , but also in each of the arguments P and Q . Moreover the boundary homomorphism (3.7.5.4) deduced from the exact sequence of the Ext^i 's attached to (3.7.5.2), through 3.6.5 and 3.7.4, is the boundary homomorphism of the theory of left exact cofibred categories over $\mathcal{C} = \text{GRAB}(\mathcal{T})^\circ$ recalled in 3.3.6, applied to the cofibred category of biextensions of (P, Q) by G , where Q and G are fixed and P is variable, up to the sign convention of 3.4.9.

We sketch the proof. By 3.6.4 and the compatibility (3.7.2), biextensions of (P, Q) by G are interpreted as objects of $\Psi_{L(P, Q)}(G)$. Consider the morphism

$$L(P, Q) = L(P) \otimes L(Q) \longrightarrow P \otimes L(Q)$$

induced by the augmentation $L(P) \rightarrow P$. From 3.5.3 and the flatness of $L(Q)$ this morphism induces isomorphisms on H_0 and H_1 , hence by (3.4.7) an equivalence

$$\Psi_{P \otimes L(Q)} \xrightarrow{\sim} \Psi_{L(P, Q)}.$$

Thus biextensions of (P, Q) by G may be interpreted as objects of $\Psi_{P \otimes L(Q)}(G)$. The exact sequence (3.7.5.2) then gives an exact sequence of chain complexes

$$0 \longrightarrow P' \otimes L(Q) \longrightarrow P \otimes L(Q) \longrightarrow P'' \otimes L(Q) \longrightarrow 0. \quad (3.7.6.1)$$

The asserted left exactness follows from 3.4.8 applied to this exact sequence. The direct verification is immediate. Finally, the homomorphism (3.7.5.4) is the boundary $\text{Ext}^0 \rightarrow \text{Ext}^1$ obtained from this exact sequence, and the last assertion is therefore a special case of 3.4.9.

3.8. Biextensions of (P, Q) trivialized on extensions of P and Q

Consider two exact sequences of abelian Groups

$$0 \longrightarrow L_1 \xrightarrow{i} L_0 \xrightarrow{u} P \longrightarrow 0, \quad 0 \longrightarrow M_1 \xrightarrow{j} M_0 \xrightarrow{v} Q \longrightarrow 0. \quad (3.8.1)$$

We wish to describe the category of biextensions E of (P, Q) by an abelian Group G whose inverse image $(u, v)^*(E)$ on (L_0, M_0) is trivial. Thus one is given a homomorphism s making the triangle

$$\begin{array}{ccc} E & & \\ \downarrow & \searrow s & \\ P \times Q & \xleftarrow{u \times v} & L_0 \times M_0 \end{array} \quad (3.8.2)$$

commute, and satisfying the additivity condition in each of the two arguments. Such an s identifies $(u, v)^*(E)$ with the trivial biextension $E_0 = L_0 \times M_0 \times G$ of (L_0, M_0) by G .

By the left exactness in P, Q of the cofibred category of biextensions of (P, Q) by a fixed G (3.7.6), the datum of E together with the trivialization s of its inverse image is essentially equivalent to the datum of two partial trivializations of the trivial biextension E_0 of (L_0, M_0) by G , on (L_1, M_0) and (L_0, M_1) , respectively, coinciding on (L_1, M_1) . Since the biextensions induced by E_0 on (L_1, M_0) , (L_0, M_1) , and (L_1, M_1) are trivial, such partial trivializations are equivalent to homomorphisms, or pairings,

$$f : L_1 \otimes M_0 \longrightarrow G, \quad g : L_0 \otimes M_1 \longrightarrow G, \quad (3.8.3)$$

and the compatibility condition says simply that these two homomorphisms coincide on $L_1 \otimes M_1$:

$$f(x_1 \otimes y_1) = g(x_1 \otimes y_1). \quad (3.8.4)$$

In terms of the homomorphism s of (3.8.2), the forms f, g are characterized by

$$\begin{aligned} s(x_1, y_0) &= f(x_1 \otimes y_0) e_{E/P(y_0)}, \\ s(x_0, y_1) &= g(x_0 \otimes y_1) e_{E/P(x_0)}. \end{aligned} \quad (3.8.5)$$

If another trivialization s' is used, then

$$s'(x_0, y_0) = s(x_0, y_0) + h(x_0 \otimes y_0) \quad (3.8.6)$$

for an arbitrary homomorphism $h : L_0 \otimes M_0 \rightarrow G$, and the two forms change to

$$\begin{aligned} f'(x_1 \otimes y_0) &= f(x_1 \otimes y_0) + h(x_1 \otimes y_0), \\ g'(x_0 \otimes y_1) &= g(x_0 \otimes y_1) + h(x_0 \otimes y_1). \end{aligned} \quad (3.8.7)$$

Thus one obtains the following description.

Proposition 3.8.8. The subgroup of $\text{Biext}^1(P, Q; G)$ formed by the classes of biextensions whose inverse image by (u, v) is a trivial biextension of (L_0, M_0) - equivalently, the kernel of

$$\text{Biext}^1(P, Q; G) \longrightarrow \text{Biext}^1(L_0, M_0; G)$$

- is canonically isomorphic, by the preceding description, to the quotient of the group of pairs of forms (f, g) as in (3.8.3), satisfying (3.8.4), by the subgroup consisting of those pairs obtained by restricting a form $h : L_0 \otimes M_0 \rightarrow G$.

We now interpret this quotient homologically, taking into account the canonical isomorphism $\text{Biext}^1(P, Q; G) \simeq \text{Ext}^1(P \otimes^L Q, G)$ of 3.6.5. Regard the data (3.8.1) as two resolution complexes of P and Q :

$$L_\bullet : 0 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \quad M_\bullet : 0 \longrightarrow M_1 \longrightarrow M_0 \longrightarrow 0. \quad (3.8.9)$$

Their tensor product is the complex

$$0 \longrightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \longrightarrow 0.$$

There is a canonical morphism in the derived category

$$P \otimes^L Q \longrightarrow L_\bullet \otimes M_\bullet,$$

hence a canonical homomorphism

$$H^1(\text{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)) \longrightarrow \text{Ext}^1(P \otimes^L Q, G). \quad (3.8.10)$$

The group of 1-cocycles of the complex $\text{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)$ identifies with the group of pairs (f, g) as in (3.8.3) satisfying (3.8.4), by assigning to such a pair the homomorphism

$$F : L_1 \otimes M_0 + L_0 \otimes M_1 \longrightarrow G, \quad F(x_1 \otimes y_0 + x_0 \otimes y_1) = f(x_1 \otimes y_0) + g(x_0 \otimes y_1). \quad (3.8.11)$$

Moreover, the coboundary of a 0-cochain $h : L_0 \otimes M_0 \rightarrow G$ is the pair of its restrictions to $L_1 \otimes M_0$ and $L_0 \otimes M_1$. Therefore the quotient described in Proposition 3.8.8 is canonically the first member of (3.8.10).

Proposition 3.8.12. Let $\xi \in \text{Ker}(\text{Biext}^1(P, Q; G) \rightarrow \text{Biext}^1(L_0, M_0; G))$ be described by a pair (f, g) as in 3.8.8. Under the isomorphism (3.6.5.1), the image of ξ in $\text{Ext}^1(P \otimes^L Q, G)$ is the image by (3.8.10) of the class represented by the cocycle (3.8.11).

The verification of this compatibility is left to the reader.

We shall use this especially with

$$L_0 = \mathbb{Z}[P], \quad M_0 = \mathbb{Z}[Q],$$

where L_1 and M_1 are the kernels of the canonical homomorphisms $\mathbb{Z}[P] \rightarrow P$ and $\mathbb{Z}[Q] \rightarrow Q$. There are canonical isomorphisms

$$L_0 \otimes^L M_0 \simeq L_0 \otimes M_0 \simeq \mathbb{Z}[P \times Q]$$

since L_0, M_0 are flat, and therefore, by (1.4.5), canonical isomorphisms

$$\text{Ext}^i(L_0 \otimes^L M_0, G) \simeq H^i(P \times Q, G_{P \times Q}).$$

Consequently, the functor sending a biextension of (L_0, M_0) by G to the torsor which it induces over $P \times Q$ is an equivalence of categories. Thus, for a biextension E of (P, Q) by G , giving a trivialization of its inverse image on (L_0, M_0) is the same as giving a section of E as an object over $P \times Q$, i.e. a trivialization of E as a torsor under $G_{P \times Q}$.

Biextensions endowed with this supplementary structure are therefore described by pairs (f, g) satisfying the compatibility (3.8.4), modulo the relation described in 3.8.8. With the notation of 3.5, one may view L_1 as the cokernel of $L_2(P) \rightarrow L_1(P)$, and similarly for M_1 . This expresses (f, g) as a system of homomorphisms

$$\begin{cases} f' : L_1(P) \otimes L_0(Q) = \mathbb{Z}(P \times P \times Q) \longrightarrow G, \\ g' : L_0(P) \otimes L_1(Q) = \mathbb{Z}(P \times Q \times Q) \longrightarrow G, \end{cases}$$

i.e. as homomorphisms of sheaves of sets

$$P \times P \times Q \longrightarrow G, \quad P \times Q \times Q \longrightarrow G,$$

satisfying five compatibility conditions: four expressing the factorization through f and g , and the condition (3.8.4). These f' and g' are precisely the partial multiplication data defining a biextension, and the five conditions are the five axioms of biextensions considered in 2.1.

Bibliography for Exposé VII

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Exposé VIII. Complements on biextensions; general properties of biextensions of group schemes

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Summary

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1. Various special cases on an arbitrary topos.
2. Pairings defined by a biextension.
3. Biextensions of group schemes (P, Q) by $\mathbb{G}_m$: generalities.
4. Extensions and biextensions of smooth connected group schemes over a field.
5. Extensions and biextensions by constant torsion-free group schemes.
6. Extensions and biextensions by the Néron model of $\mathbb{G}_m$.
7. Canonical prolongations of extensions and biextensions by $\mathbb{G}_m$.

0. Introduction

We continue here the general study of biextensions begun in the preceding exposé. We first give complements in the case of an arbitrary base topos, developing in particular the formalism of the pairings attached to biextensions, whose importance in the theory of abelian schemes is well known. We then begin the study of biextensions of general group schemes, the most interesting case for us being biextensions by the multiplicative group $\mathbb{G}_m$. As a technical intermediate step we shall also have to study biextensions by other types of group schemes.

The most important result for later use is 7.1. It gives conditions under which, for P and Q smooth over the generic point η of a trait, a biextension of (P_η, Q_η) by $\mathbb{G}_m$ extends to a biextension of (P, Q) by $\mathbb{G}_m$, determined up to unique isomorphism. This will be convenient in the next exposé, which concerns Néron models of abelian varieties. It is likely that these developments will be useful elsewhere as well, although the later part of the seminar will use them only sparingly.

When working with group schemes over a scheme S , we shall tacitly use the fppf topos of S , attached to the $\mathcal{U}$ -site of schemes locally of finite presentation over S , endowed with the fppf topology. Most statements would remain valid, and essentially equivalent, for other topologies whose underlying category contains the schemes locally of finite presentation over S , such as the Zariski, étale, or fpqc topology. The reason is descent: the category of extensions of a group scheme P , locally of finite presentation over S , by $\mathbb{G}_m$, and similarly the notion of biextension, do not depend, up to equivalence, on this choice. Technically, the fppf topology, or a finer topology such as fpqc, is preferred because the homomorphism $n \cdot \text{id}$ of $\mathbb{G}_m$ is an fppf epimorphism for every $n > 0$, whereas it is not in general a Zariski or étale epimorphism.

Finally, most statements on extensions or biextensions by $\mathbb{G}_m$ remain true if $\mathbb{G}_m$ is replaced by a torus G . Since such a G is locally isomorphic, for the fppf and even for the étale topology, to $\mathbb{G}_m^r$, this is essentially a trivial generalization.

1. Various special cases on an arbitrary topos

As in Exposé VII, let $\mathcal{T}$ be a topos, and let P, Q, G be three abelian Groups of $\mathcal{T}$. We use the canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^L Q, G). \quad (1.1.1)$$

Recall also the Cartan isomorphisms

$$\mathrm{Ext}^1(P \otimes^L Q, G) \simeq \mathrm{Ext}^1(P, \mathrm{RHom}(Q, G)) \simeq \mathrm{Ext}^1(Q, \mathrm{RHom}(P, G)), \quad (1.1.2)$$

functorial in the three arguments. From (1.1.1) one obtains the exact sequence

$$\begin{aligned} 0 \rightarrow \mathrm{Ext}^1(P \otimes Q, G) \rightarrow \mathrm{Biext}^1(P, Q; G) \rightarrow \mathrm{Hom}(\mathrm{Tor}_1(P, Q), G) \rightarrow \\ \mathrm{Ext}^2(P \otimes Q, G) \rightarrow \mathrm{Ext}^2(P \otimes^L Q, G) \rightarrow \cdots, \end{aligned} \quad (1.1.3)$$

because the homology objects of $P \otimes^L Q$ are the $\mathrm{Tor}_i(P, Q)$, zero for $i \geq 1$. Similarly, since the homology objects of $\mathrm{RHom}(Q, G)$ are the $\mathrm{Ext}^i(Q, G)$, the first isomorphism in (1.1.2) gives a five-term exact sequence

$$\begin{aligned} 0 \rightarrow \mathrm{Ext}^1(P, \mathrm{Hom}(Q, G)) \rightarrow \mathrm{Biext}^1(P, Q; G) \rightarrow \mathrm{Hom}(P, \mathrm{Ext}^1(Q, G)) \rightarrow \\ \mathrm{Ext}^2(P, \mathrm{Hom}(Q, G)) \rightarrow \mathrm{Ext}^2(P, \mathrm{RHom}(Q, G)). \end{aligned} \quad (1.1.4)$$

This sequence is associated with the usual spectral sequence

$$\mathrm{Ext}^*(P, \mathrm{RHom}(Q, G)) \longleftarrow \mathrm{Ext}^p(P, \mathrm{Ext}^q(Q, G)).$$

A fully faithful functor may be used to describe the first arrow in (1.1.3), respectively in (1.1.4), namely

$$\mathrm{EXT}(P \otimes Q, G) \rightarrow \mathrm{BIEXT}(P, Q; G), \quad (1.1.5)$$

or

$$\mathrm{EXT}(P, \mathrm{Hom}(Q, G)) \rightarrow \mathrm{BIEXT}(P, Q; G). \quad (1.1.6)$$

The symmetric Cartan isomorphism yields the analogous functor

$$\mathrm{EXT}(Q, \mathrm{Hom}(P, G)) \rightarrow \mathrm{BIEXT}(P, Q; G). \quad (1.1.7)$$

If $\mathrm{Tor}_1(P, Q) = 0$, then (1.1.3) gives a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes Q, G). \quad (1.2.2)$$

This is the case, for example, if P or Q is a flat $\mathbb{Z}$ -Module; more precisely, under the same hypothesis the functor (1.1.5) is an equivalence of categories.

If $P \otimes Q = 0$, then (1.1.3) gives a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}(\mathrm{Tor}_1(P, Q), G). \quad (1.3.2)$$

A particularly useful case occurs when there is an integer $n > 0$ such that

$$nP = P, \quad Q = \varinjlim_i n^i Q, \quad (1.3.3)$$

that is, P is divisible by n and Q is a filtered limit of subgroups killed by powers of n . Put

$$T_n(P) = ({}_n P)_{i \geq 0}, \quad n^\infty P = \varinjlim_i n^i P.$$

Then

$$P \otimes Q = 0, \quad \mathrm{Tor}_1(P, Q) \simeq T_n(P) \otimes Q \simeq {}_{n^\infty}P \otimes T_n(Q), \quad (1.3.4)$$

and hence

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}(T_n(P), D(Q)) \simeq \mathrm{Hom}(Q, D(T_n(P))). \quad (1.3.5)$$

Here D denotes $\mathrm{Hom}(-, G)$. The transition map in the inductive system appearing in (1.3.4) is obtained by identifying the first term with ${}_n P \otimes {}_n Q$, using n^{j-i} id on the first factor and the inclusion on the second. For a group M killed by n^i one defines

$$T_n(P) \otimes M \text{ dfn } {}_n P \otimes M,$$

and $D(Q) = (\mathrm{Hom}({}_n Q, G))_{i \geq 0}$. The remaining identifications are those of morphisms of projective systems.

The calculation of $\mathrm{Tor}_1(P, Q)$ used above is reduced to the case in which Q is killed by n^i . It suffices to use the exact sequence

$$0 \rightarrow P \xrightarrow{m} P \rightarrow P/mP \rightarrow 0$$

and the standard connecting morphism to get, when $mP = P$ and $mQ = 0$,

$$\mathrm{Tor}_1(P, Q) \xleftarrow{\sim} {}_m P \otimes_{\mathbb{Z}/m\mathbb{Z}} Q. \quad (1.3.6)$$

The choice of this isomorphism amounts only to a sign convention.

If

$$\mathrm{Hom}(Q, G) = 0, \quad (1.4.1)$$

then (1.1.4) gives a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}(P, \mathrm{Ext}^1(Q, G)). \quad (1.4.2)$$

This case occurs, for example, when Q is an abelian scheme and $G = \mathbb{G}_m$. Then the functor in P , $P \mapsto \mathrm{Biext}^1(P, Q; G)$, is representable by

$$Q^* = \mathrm{Ext}^1(Q, G),$$

and one obtains a canonical biextension E^Q of (Q^*, Q) by G , uniquely determined up to unique isomorphism. This biextension, or the symmetric biextension ${}^s E^Q$ of (Q, Q^*) by G , may be called the Weil biextension of (Q^*, Q) , or of (Q, Q^*) , by G .

If

$$\mathrm{Hom}(P, \mathrm{Ext}^1(Q, G)) = 0, \quad (1.5.1)$$

then (1.1.4) yields the canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P, \mathrm{Hom}(Q, G)). \quad (1.5.2)$$

More precisely, the fully faithful functor (1.1.6) is then an equivalence of categories. If both

$$\mathrm{Hom}(P, G) = 0, \quad \mathrm{Hom}(P, \mathrm{Ext}^1(Q, G)) = 0 \quad (1.6.1)$$

hold, then the two isomorphisms of 1.4 combine into a duality formula

$$\mathrm{Ext}^1(P, \mathrm{Hom}(Q, G)) \simeq \mathrm{Hom}(Q, \mathrm{Ext}^1(P, G)) \simeq \mathrm{Biext}^1(P, Q; G). \quad (1.6.2)$$

The first condition in (1.6.1) implies $\mathrm{Hom}(P \otimes Q, G) = 0$. Hence the category of commutative extensions of P by $\mathrm{Hom}(Q, G)$ is rigid, and the same is true of $\mathrm{BIEXT}(P, Q; G)$. Therefore their

structure is known once the group of isomorphism classes is known; by (1.6.2) this group is canonically $\text{Hom}(Q, \text{Ext}^1(P, G))$.

The map (1.6.2) is actually defined without assuming (1.6.1). It may be described directly, without passing through $\text{Biext}^1(P, Q; G)$, in terms of the natural pairing

$$Q \times Q' \longrightarrow G, \quad Q' = \text{Hom}(Q, G),$$

which is equivalently a homomorphism $Q \rightarrow \text{Hom}(Q', G)$. It gives the composite

$$\text{Ext}^1(P, Q') \times Q \longrightarrow \text{Ext}^1(P, Q') \times \text{Hom}(Q', G) \longrightarrow \text{Ext}^1(P, G), \quad (1.6.3)$$

recovering (1.6.2).

2. Pairings defined by a biextension

Let P, Q be two abelian Groups of $\mathcal{T}$, and let $n > 0$ be an integer. We write ${}_nP$ for the kernel of multiplication by n on P . We define a canonical homomorphism

$$\alpha = \alpha_{P,Q}^n : {}_nP \otimes {}_nQ \longrightarrow \text{Tor}_1(P, Q). \quad (2.1.1)$$

It is the composite

$${}_nP \otimes {}_nQ \longrightarrow \text{Tor}_1({}_nP, {}_nQ) \longrightarrow \text{Tor}_1(P, Q), \quad (2.1.2)$$

where the first arrow is defined first in the case in which P and Q are killed by n . Suppose for instance that ${}_nQ = Q$. Taking a flat resolution $L_\bullet$ of P , one has canonical isomorphisms

$$P \otimes_{\mathbb{Z}}^L Q \simeq L_\bullet \otimes Q \simeq (L_\bullet \otimes \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} Q,$$

and this gives a homological Kunneth spectral sequence

$$\text{Tor}_*^{\mathbb{Z}}(P, Q) \longleftarrow E_{p,q}^2 = \text{Tor}_p^{\mathbb{Z}/n\mathbb{Z}}(\text{Tor}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}), Q). \quad (2.1.4)$$

The usual flat resolution

$$0 \longrightarrow \mathbb{Z}_{\mathcal{T}} \xrightarrow{n} \mathbb{Z}_{\mathcal{T}} \longrightarrow 0$$

of $(\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$ gives

$$\text{Tor}_0^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) = P_n = P/nP, \quad \text{Tor}_1^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) = {}_nP,$$

and higher Tor_i 's vanish. Therefore (2.1.4) reduces essentially to the exact sequence

$$0 \rightarrow \text{Tor}_2^{\mathbb{Z}/n\mathbb{Z}}(P_n, Q) \rightarrow {}_nP \otimes Q \xrightarrow{\alpha} \text{Tor}_1^{\mathbb{Z}}(P, Q) \rightarrow \text{Tor}_1^{\mathbb{Z}/n\mathbb{Z}}(P_n, Q) \rightarrow \cdots \quad (2.1.5)$$

This defines the desired map and shows that it is functorial in P and Q . In particular it is an isomorphism when ${}_nP$ is a flat $\mathbb{Z}/n\mathbb{Z}$ -Module (for example when $nP = P$ and ${}_nP$ is flat), or when Q is flat over $\mathbb{Z}/n\mathbb{Z}$.

The map α may be described more explicitly. To know $\alpha(p, q)$ for sections $p \in {}_nP$ and $q \in {}_nQ$, one reduces by functoriality to the case $Q = (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$ and $q = 1$. In that case $\text{Tor}_1^{\mathbb{Z}}(P, Q) \simeq {}_nP$, and α is the identity. Equivalently, one further reduces to $P = Q = (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$, where

$$(\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}} \xrightarrow{\sim} \text{Tor}_1^{\mathbb{Z}}((\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}, (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}) \quad (2.1.8.1)$$

is induced by the canonical flat resolution

$$0 \longrightarrow \mathbb{Z}_{\mathcal{T}} \xrightarrow{n} \mathbb{Z}_{\mathcal{T}} \longrightarrow 0 \quad (2.1.8.2)$$

of the first factor.

The definition of α is not symmetric. By exchanging P and Q in the construction, one obtains an analogous map

$$\beta_{P,Q}^n : {}_n P \otimes {}_n Q \longrightarrow \mathrm{Tor}_1(P, Q). \quad (2.1.9.1)$$

The two maps may also be characterized by the commutative square

$$\begin{array}{ccc} {}_n P \otimes {}_n Q & \xrightarrow{\sim} & {}_n Q \otimes {}_n P \\ \beta_{P,Q}^n \downarrow & & \downarrow \alpha_{Q,P}^n \\ \mathrm{Tor}_1(P, Q) & \xrightarrow{\sim} & \mathrm{Tor}_1(Q, P), \end{array} \quad (2.1.9.2)$$

where the horizontal arrows are the symmetry isomorphisms. Their relation is

$$\beta_{P,Q}^n = -\alpha_{P,Q}^n. \quad (2.1.9.3)$$

Lemma 2.1.10. Let P, Q be two commutative Groups of $\mathcal{T}$, and let $n > 0$. Then the square

$$\begin{array}{ccc} {}_n P \otimes {}_n Q & \xrightarrow{\sim} & {}_n Q \otimes {}_n P \\ \alpha_{P,Q}^n \downarrow & & \downarrow \alpha_{Q,P}^n \\ \mathrm{Tor}_1(P, Q) & \xrightarrow{\sim} & \mathrm{Tor}_1(Q, P) \end{array} \quad (2.1.10.1)$$

is anti-commutative, the horizontal arrows being the symmetry isomorphisms.

Proof. The procedure of 2.1.8 reduces the question to the case $P = Q = (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$. It is enough then to work in the punctual topos, i.e. in ordinary commutative groups. Take

$$L_{\bullet} : 0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0$$

as a flat resolution of $P = Q = \mathbb{Z}/n\mathbb{Z}$. Using $L_{\bullet} \otimes L_{\bullet}$, the isomorphism defining α is represented by the cycle

$$-1 \otimes l_1 + l_1 \otimes 1,$$

whereas the isomorphism defining β is represented by

$$1 \otimes l_1 - l_1 \otimes 1.$$

This proves the lemma.